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STORM-PhysNet: A Multi-Horizon Transformer for Geostationary Relativistic Electron Flux Forecasting with Physics-Inspired Components and Cross-Satellite Transfer

SAMARTH BN[1], (Student Member, IEEE)
[1]RV College of Engineering, Bengaluru, India (e-mail: samarthbn.ec25@rvce.edu.in)

Corresponding author: Samarth BN (e-mail: samarthbn.ec25@rvce.edu.in).

This work extends a concurrent conference submission on the same system. The present manuscript adds the wider-delay ablation, the bagged-Transformer control, the architecture-matched Transformer control, expanded interpretability diagnostics, and the full multi-seed GRASP transfer analysis.

ABSTRACT Energetic electrons above 2 MeV at geostationary Earth orbit (GEO) are a persistent internal-charging hazard for satellites. We present STORM-PhysNet, a multi-horizon forecast model that couples a standard Transformer encoder over a 72-hour window with a learnable L1–Earth propagation delay, a B_z-conditioned physics gate, and residual heads for simultaneous 1-hour, 6-hour, and 12-hour log-flux forecasts. On hourly GOES–OMNI data with fifteen random initializations of the same chronological split, STORM-Bz improves short-horizon reliability ($\mathrm{PE}_{1\mathrm{h}} = 0.986$ versus Transformer 0.977, paired $p = 0.002$) and remains clearly positive against persistence (PE≈ 0.32 versus ≈ -0.14). At 6 h and 12 h a plain Transformer remains competitive or slightly stronger as a single model. A validation-selected linear ensemble ($\alpha^* = 0.3$) and seed-bagged STORM both reach 6 h PE of 0.911. An architecture-matched Transformer control ($d_{\mathrm{model}} = 128$, two layers, four heads) reached mean $\mathrm{PE}_{1\mathrm{h}} = 0.981$ and $\mathrm{PE}_{6\mathrm{h}} = 0.907$ across fifteen seeds; bagging those seeds yields 0.986 / 0.919 / 0.877, establishing the bagged architecture-matched Transformer as the strongest model at the 6 h and 12 h horizons. While STORM dominates the single-model 1-hour forecast (0.988 bagged), encoder width/depth and bagging move multi-horizon PE more than the physics modules at inference. Controlled ablations show that the short-horizon gain is a property of the trained package as a whole rather than of any individual physics module at inference.
Under synthetic solar-wind input noise and targeted channel dropout, STORM degrades more slowly than the Transformer, especially at 1-hour. On GSAT-19 GRASP at Indian longitude, zero-shot transfer is weak at long horizons; fine-tuning on GRASP raises 6 h PE from 0.449 to 0.599 (paired $p = 7.6 \times 10^{-7}$) and 12 h PE from 0.182 to 0.517 ($p = 3.6 \times 10^{-10}$) across fifteen seeds. We report bootstrap confidence intervals, paired tests, case studies, and release evaluation scripts so that later work can reuse the same splits and PE definitions.

INDEX TERMS Geostationary orbit, GRASP, physics-inspired neural network, prediction efficiency, radiation belt, relativistic electrons, space weather, Transformer encoder, transfer learning

I. INTRODUCTION

THE outer electron radiation belt at geostationary Earth orbit (GEO) is one of the most dynamically variable plasma environments in near-Earth space. Relativistic electrons with energies $E > 2\,\mathrm{MeV}$ can undergo flux variations spanning four to five orders of magnitude over timescales of hours to days, driven by the interplay of solar wind forcing, magnetospheric wave-particle interactions, and global current systems [1]–[3]. These populations represent a chronic engineering hazard: energetic electrons penetrating satellite shielding accumulate charge on internal dielectrics and conductors, potentially triggering electrostatic discharges that damage or destroy spacecraft electronics [1]. With the rapid expansion of the operational GEO belt—hosting communication, broadcasting, and navigation assets—accurate forecasting of electron fluxes has become an operational priority for satellite operators and space agencies worldwide.

Forecast models for GEO electrons have evolved considerably over four decades. The earliest approaches used linear autoregressive filters and empirical correlations with solar wind speed and geomagnetic indices [4], [5]. NOAA's Relativistic Electron Forecast Model (REFM), which has operated since the 1990s, remains a reference benchmark for the community [5]. We do not include REFM as a numeric baseline because it targets daily integral fluence rather than hourly log-flux; a direct PE comparison would be misleading [5]. The subsequent generation of feed-forward neural networks demonstrated that nonlinear mappings from solar wind and IMF parameters could improve upon these linear baselines [7], [8]. LSTM-based architectures then raised the bar further by explicitly capturing multi-day temporal dependencies in the solar wind driving [9]–[12]. More recently, Transformer and attention-based models—originally developed for natural language processing [13]—have been adapted for space weather time series, reporting daily-fluence prediction efficiencies of 0.85–0.92 with or without pre-training strategies [14]–[16].

Despite this rapid progress, several gaps remain. First, most published models target either a single forecast horizon (typically 24 h fluence) or a small number of multi-hour lead times; the short-horizon regime (about 1 h) that is critical for immediate operational response has received less attention as a stand-alone multi-seed evaluation target. Second, the community lacks a controlled multi-seed benchmark that directly compares physics-inspired and physics-agnostic architectures at equivalent training budgets on the same chronological splits. Third, nearly all published work focuses on GOES satellites at American/European longitudes; the performance of transfer strategies for Indian-longitude GEO assets such as ISRO's GSAT-19—which carries the Geostationary Radiation Alert and Monitoring System (GRASP) instrument—has not been systematically characterized.

STORM-PhysNet is designed around these three gaps.

A. CONTRIBUTIONS

- A Transformer encoder with adaptive L1–Earth delay and a B_z-conditioned gate, evaluated at 1 h, 6 h, and 12 h under fifteen matched seeds and chronological splits.
- Evidence that STORM improves short-horizon reliability relative to a strong Transformer (significant $\mathrm{PE}_{\mathrm{clim}}$ gain at 1 h; large $\mathrm{PE}_{\mathrm{pers}}$ gap). A linear ensemble with validation-selected $\alpha^* = 0.3$ further improves multi-horizon PE relative to either model alone.
- Seed-bagged STORM as a pure-STORM multi-horizon alternative that approaches ensemble skill without a second architecture at inference.
- Noise and missing-channel robustness comparisons, GRASP zero-shot versus fine-tune transfer with fifteen seeds, and a null ablation showing that short-horizon-only physics losses do not move PE.
- Case studies, delay/gate diagnostics, and released evaluation scripts for reuse of the same PE definitions and splits.

II. RELATED WORK

A. PHYSICS OF OUTER RADIATION BELT ELECTRONS AT GEO

The outer radiation belt at GEO ($L \approx 6.6$) is maintained by a dynamic balance between electron acceleration and loss. Storm-time acceleration and loss are reviewed in detail by Reeves et al. [17] and Thorne [18]. Which storms produce large GEO electron responses is discussed by O'Brien et al. [19]; broader source–loss modeling is reviewed by Shprits et al. [20]. Acceleration occurs primarily through whistler-mode chorus wave interactions near the magnetic equator [2], [3], [6]. Ultra-low frequency (ULF) waves additionally drive inward radial diffusion. Loss processes include precipitation into the atmosphere and magnetopause shadowing during storm compressions. The net response of GEO flux to a geomagnetic storm depends on the relative balance of these mechanisms and the direction and magnitude of the interplanetary magnetic field B_z [2]. This motivates our choice to condition the primary gate module on B_z.

B. MACHINE LEARNING FOR GEO ELECTRON FLUX

Camporeale's review framed both the opportunities and the methodological pitfalls of data-driven space weather modeling [21]. Morley et al. defined and advocated for prediction efficiency as the preferred scalar skill score for relativistic electron forecasts [22].

Early neural network models established that solar wind speed, IMF B_z, and geomagnetic indices carry predictive signal for GEO electrons [7]. Wei et al. demonstrated that a deep network on multi-day solar wind histories achieves strong daily PE [8]. Son et al. pushed to 72-hour multi-step hourly forecasts with LSTM [9]. Zhang et al. combined empirical mode decomposition with LSTM [10]. Simms et al. focused on classification of enhancement events [11].

Transformer-based architectures entered the GEO flux literature around 2023–2024 [14]–[16]. Stacking ensembles have also been explored for short-horizon energetic electron prediction [23]. The iTransformer of Liu et al. [24] was briefly evaluated as an alternative backbone; under the same multi-seed protocol it showed higher variance across seeds than the standard temporal-attention Transformer, which was therefore retained for all reported experiments.

Beyond GEO, Sinha et al. developed PreMevE [25]; Botek et al. [26] applied neural networks at low Earth orbit; Wang et al. [12] used LSTM with MagEIS observations.

A recurring theme in this literature is that headline PE depends strongly on the chosen cadence, energy channel, solar-cycle phase, and persistence definition. Daily fluence

TABLE 1. Design comparison with closely related GEO electron flux forecasting studies. PE numbers are omitted on purpose: they are not interchangeable across different periods, cadences, and baselines [21], [22].

Study	Core model	Horizon focus	Relative to this work
Wei et al. (2018)	LSTM	1-day fluence	Daily target; no multi-seed protocol
Son et al. (2022)	MLP	1–72 h hourly	No physics modules; no cross-satellite transfer
Zhang et al. (2022)	EMD-LSTM	1-day	Decomposition pipeline; daily resolution
Tan et al. (2024)	TPA-LSTM / TF	Up to 3 days (5-min)	Multi-energy channels; no delay/gate ablations
Sun et al. (2024)	LSTM / TF	1 h–1 day	Closest multi-horizon TF baseline; no module ablations or GRASP transfer
Sun et al. (2025)	TF + data strategies	Daily fluence	Fluence target; few-shot/pretrain focus
Sinha et al. (2021)	PreMevE ensemble	∼1–2 day belt-wide	Multi-mission belt model; not GEO-only multi-horizon heads
This work	TF + delay + B_z gate	1 h / 6 h / 12 h	15-seed protocol; honest module ablations; GOES→GRASP transfer

scores near 0.9 are not directly comparable to hourly log-flux PE on a different year range [21], [22]. Our evaluation therefore emphasizes internal controls: the same chronological splits, the same fifteen seeds, and the same PE formula for every architecture, including ablations. That design does not replace a community benchmark, but it does make relative statements among the models in this paper interpretable.

Physics-inspired machine learning for radiation-belt forecasting is still less common than pure sequence models. Where physics appears, it is often encoded as post-hoc diagnostics, hard-wired lag features, or loss penalties rather than as trainable modules whose outputs can be inspected (for example, a learned transit delay whose distribution can be compared to the L1–Earth travel-time window). STORM-PhysNet is deliberately conservative on the encoder side—a standard temporal Transformer—so that interpretability claims attach to the delay and gate, not to an exotic backbone.

C. SOLAR WIND PROPAGATION DELAY

Solar-wind transit from L1 to the magnetopause is typically on the order of 0.5–1.5 h [27] depending on solar-wind speed. The delay module is constrained to $[0.5, 1.5]$ h, a band chosen to cover that range while still allowing limited storm-time deviation without breaking causality. We do not claim a unique recovered constant of 1.09 h from every window.

D. TRANSFER LEARNING AND CROSS-SATELLITE ADAPTATION

Most radiation-belt transfer efforts have focused on cross-calibration of flux measurements across fleets [25]. The specific problem of adapting a model trained on GOES-15 to GSAT-19 GRASP in a different longitude sector has not been previously reported in the form studied here. We explore transfer by adapting a GOES-pretrained model to GRASP with limited local labels.

III. DATA AND PROBLEM SETUP

A. GOES SATELLITE DATA

We use measurements of $E > 2\,\mathrm{MeV}$ integral electron flux from the GOES-15 satellite. The chronological working sample spans exactly January 1, 2012 to December 31, 2016. Raw one-minute flux values are aggregated to hourly means; the within-hour standard deviation is retained as an input feature so that intra-hour variability is not discarded. All flux values are log-transformed ($\log_{10}$ of particle cm^{-2}s^{-1}sr^{-1}).

B. OMNI SOLAR-WIND AND GEOMAGNETIC INPUTS

Solar wind and interplanetary magnetic field parameters are drawn from the OMNI 1-hour dataset. The released dataloader uses sixteen solar-wind / geomagnetic channels plus log-flux history (17 inputs total after concatenation):

- Plasma and pressure: V_{sw}, proton density, P_{dyn}
- Magnetic field: B_t, B_z
- Derived activity: B_z-south duration, storm-onset hours, $B_z \times V_{sw}$
- Rolling moments: V_{sw} at 1/6/24 h, B_z at 1/6 h
- Indices: Kp, Dst
- Within-hour variability: $\mathrm{std}(\log \mathrm{flux})$ over the 1-minute samples

Forecast targets use integer hourly leads of 1 h, 6 h, and 12 h on the hourly GOES series (config/dataloader: $[1, 6, 12]$). A 72-hour history window ($T = 72$) is used at each forecast origin.

TABLE 2. Input feature inventory used by the released dataloader

Category	Channels
Plasma & pressure	V_{sw}, density, P_{dyn}
Magnetic field	B_t, B_z
Derived drivers	B_z-south duration, storm-onset hours, $B_z \times V_{sw}$
Rolling moments	V_{sw} (1/6/24 h), B_z (1/6 h)
Geomagnetic indices	Kp, Dst
Within-hour flux variability	log-flux std over 1-min samples
Autoregressive target history	log-flux (separate input stream)

C. CHRONOLOGICAL SPLITS

Splits are purely chronological: the earliest 70% of the merged GOES–OMNI series for training, the next 15% for validation, and the final 15% for held-out testing. No shuffling is applied. Storms that straddle a boundary are assigned to the later split.

D. FORECAST TARGETS AND HORIZONS

The model simultaneously predicts log-flux at three lead times:

$$\hat{\mathbf{y}}_t = \left[\hat{y}_{t+1\mathrm{h}},\, \hat{y}_{t+6\mathrm{h}},\, \hat{y}_{t+12\mathrm{h}}\right]. \tag{1}$$

The 1 h head targets an operationally actionable warning window. The 6 h head matches the lead time most commonly benchmarked in the literature. The 12 h head tests medium-range predictability.

E. EVALUATION METRICS

We report two PE definitions so that results remain comparable both to climatological skill and to persistence-based operational baselines following Morley et al. [22] and the methodological cautions in Camporeale [21]:

$$\mathrm{PE}_{\mathrm{clim}} = 1 - \frac{\mathrm{MSE}(\hat{y}, y)}{\mathrm{Var}(y)}, \tag{2}$$

$$\mathrm{PE}_{\mathrm{pers}} = 1 - \frac{\mathrm{MSE}(\hat{y}, y)}{\mathrm{MSE}(y^{\mathrm{pers}}, y)}. \tag{3}$$

Unless stated otherwise, headline GOES tables use $\mathrm{PE}_{\mathrm{clim}}$ with fifteen seeds and bootstrap 95% confidence intervals. The fifteen seeds measure sensitivity to random initialization on a fixed chronological split; they do not constitute independent trials across different solar-cycle phases or data periods. $\mathrm{PE}_{\mathrm{pers}}$ is emphasized at the 1-hour horizon, where persistence is a strong operational baseline. Storm windows use Dst $\leq -50\,\mathrm{nT}$.

F. GRASP DATASET

GRASP aboard GSAT-19 provides GEO radiation measurements at Indian longitude (≈82°E). We use the available record covering approximately July 2017 to September 2018—an ≈14–15 month interval in the declining phase of Solar Cycle 24. After alignment, GRASP supplies fewer co-located input channels than the GOES pipeline; this feature mismatch is a proximate cause of zero-shot failure and motivates input-projection adaptation. Storm windows on the GRASP test set are defined by Dst $\leq -50\,\mathrm{nT}$ (118 of 1781 test hours, 6.6%).

IV. METHOD: STORM-PHYSNET

A. OVERALL ARCHITECTURE

STORM-PhysNet follows an encode–delay–gate–decode design (Fig. 1). For a batch of input sequences $\mathbf{X} \in \mathbb{R}^{B\times T\times F}$: (1) an input projection maps the concatenated solar-wind and flux features at each time step to dimension d_{model}; (2) the adaptive delay module interpolates the solar-wind channels by τ time steps; (3) a Transformer encoder processes the sequence and the final time-step representation is extracted; (4) the B_z-conditioned gate computes a per-channel modulation vector; (5) three independent residual heads produce the three horizon forecasts. Measured parameter counts are approximately 3.43×10^5 for STORM-Bz and 8.45×10^5 for the Transformer baseline (default hyperparameters). The two models differ in both width and depth; they are not architecture-matched. We additionally trained an architecture-matched Transformer control with the same encoder size as STORM ($d_{\mathrm{model}} = 128$, two layers, four heads; $\approx 1.19 \times 10^6$ parameters) under the same chronological split and fifteen seeds. Default-baseline PE differences should still be read with the parameter-count disparity in mind; matched-control PE is reported separately.

B. ADAPTIVE PROPAGATION DELAY

STORM-PhysNet predicts a per-window delay $\tau = f_\theta(X_{\mathrm{sw}})$ from a short conditioning network on the early part of the input window, with a global learned bias term for stable initialization, constrained to $[0.5, 1.5]$ h. Differentiable linear interpolation then time-shifts the solar-wind features:

$$\tilde{X}_{\mathrm{sw}}(t) = (1-\delta)\, X_{\mathrm{sw}}(t) + \delta\, X_{\mathrm{sw}}(t-1), \quad \delta = \tau - \lfloor \tau \rfloor. \tag{4}$$

Here δ denotes the fractional delay. Across seeds, predicted delays are constrained to the physical $[0.5, 1.5]$ h band.

C. B_Z-CONDITIONED PHYSICS GATE

Sustained southward IMF B_z triggers dayside reconnection and ring-current injection [2]. The primary gate maps a short vector of southward-B_z statistics through an MLP and a clamped sigmoid:

$$\mathbf{u}_{B_z} = [\bar{B}_{z,\mathrm{south}}, \max(B_{z,\mathrm{south}}), \Delta t_{B_z}] \tag{5}$$

$$\boldsymbol{\sigma}_{\mathrm{raw}} = \mathrm{MLP}(\mathbf{u}_{B_z}) \in (0,1)^{d_{\mathrm{model}}} \tag{6}$$

$$\mathbf{g} = g_{\min} + (1 - g_{\min})\boldsymbol{\sigma}_{\mathrm{raw}} \tag{7}$$

$$\mathbf{h}_{\mathrm{gated}} = \mathbf{g} \odot \mathbf{h} \tag{8}$$

with $g_{\min} = 0.1$ to avoid complete shutdown during quiet times.

D. TRANSFORMER BACKBONE AND RESIDUAL HEADS

We use a standard Transformer encoder [13]: each of the $T = 72$ time steps is a token; self-attention is applied across time; the final hidden state feeds the gate and heads. Residual heads project the shared latent vector to the three horizons with a skip connection from the last-observed flux value.

E. PHYSICS REGULARIZATION LOSS

Training minimises a composite objective: a primary multi-horizon residual loss (beta-NLL with a 2.5× under-prediction penalty during storm periods), plus auxiliary Dst/Kp and storm-onset terms and small regularisers (monotonicity, temporal smoothness, delay, variance calibration, and an Akasofu-ε coupling consistency term). Horizon

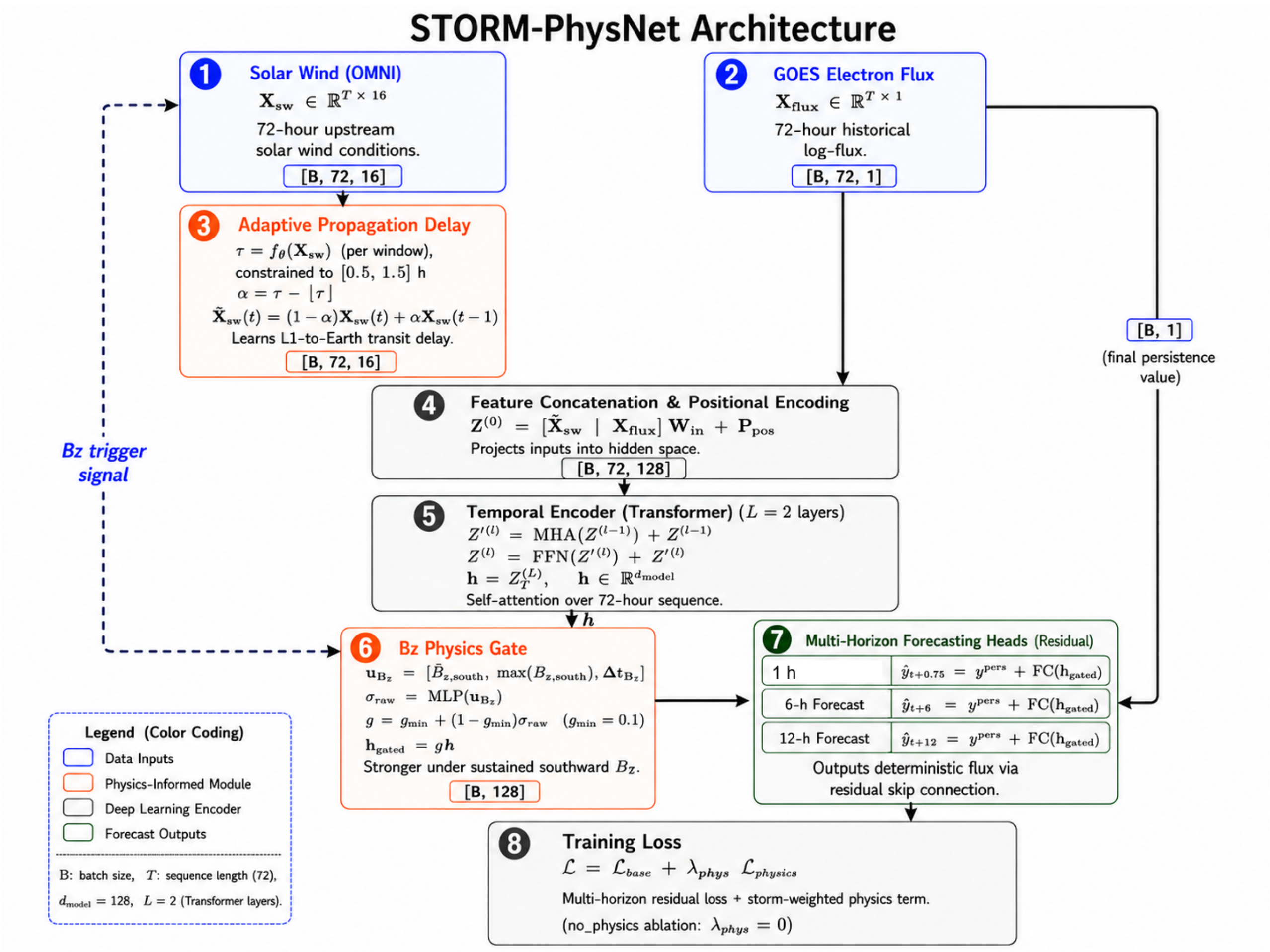

FIGURE 1. STORM-PhysNet data flow: OMNI solar wind and GOES flux enter an adaptive propagation delay and temporal Transformer encoder; a B_z-conditioned gate modulates the representation under southward IMF; three residual heads produce simultaneous 1-hour, 6-hour, and 12-hour log-flux forecasts.

weights emphasise the short horizon. Coefficient values follow the released configuration. We did not ablate every auxiliary term individually; reported PE reflects the full composite objective used in training.

F. SUPPLEMENTARY GATE NONLINEARITIES

The main STORM-Bz configuration uses a physics-motivated B_z gate whose inputs are southward IMF duration and related solar-wind features. To test whether PE depends on that specific nonlinearity—or only on the shared training package (storm-biased sampling, residual heads, multi-horizon loss)—we trained three alternative gates that keep the same Transformer backbone, delay module, residual heads, data splits, and fifteen seeds $\{42, \ldots, 56\}$. Only the gate block (and, for one variant, an auxiliary head) changes. Internal code keys differ from the names below; see the repository map in the "Repository name map" paragraph below.

a: Rectified Driver Gate (RDG).

RDG is a circuit-inspired shape prior, not a claim that GEO electrons behave like a vacuum tube. Solar-wind driving is treated as an emission-like control signal: a learned estimator of driver strength (built from v_{sw} and B_z) is passed through a rectified, smooth nonlinearity so that only one polarity of driving strongly opens the gate. The gated representation fed to the residual heads is

$$h_{\text{g}} = g_{\text{RDG}}(v_{\text{sw}}, B_z) \odot h,$$

where h is the encoder state and $g_{\text{RDG}} \in (0, 1]$ is the gate activation. Relative to the physics B_z gate, RDG emphasizes half-wave / driver-aligned modulation rather than an explicit southward-duration feature path.

b: RDG with Spectral Head (RDG-S).

RDG-S uses the same RDG block and adds a small auxiliary head that predicts spectral parameters (κ, θ) and maps them to an optional $\geq 2\,\text{MeV}$ log-flux estimate. The primary multi-horizon loss still trains the residual flux heads; the spectral branch is additive and does not replace $\hat{y}$. This tests whether an explicit spectral parameterization improves PE when the gate is already non-physics-B_z.

TABLE 3. Supplementary gates (fifteen seeds). Mean $\mathrm{PE}_{\mathrm{clim}}$ on the fixed chronological GOES test split. STORM-Bz from the main table is repeated for reference.

Gate	$\mathrm{PE}_{1\mathrm{h}}$	$\mathrm{PE}_{6\mathrm{h}}$	$\mathrm{PE}_{12\mathrm{h}}$
STORM-Bz (physics B_z)	0.986	0.897	0.851
RDG	0.987	0.910	0.868
RDG-S	0.987	0.909	0.867
SDG	0.987	0.911	0.868

c: Saturating Dose Gate (SDG).

SDG is a dose-response shape prior: recent energy-related input is accumulated with a learnable scale and passed through a saturating curve (smooth approach to a ceiling), so the gate opens more under sustained forcing and levels off rather than growing without bound. This is an accumulation-then-saturation inductive bias, not a biological model of the radiation belts. Formally, if d_t denotes a learned scalar summary of recent driving,

$$g_{\mathrm{SDG}} = g_{\min} + (1 - g_{\min})\,\sigma(\alpha d_t + \beta),$$

with learnable (α, β) and a floor $g_{\min}$ to avoid hard zeros.

d: Protocol and results.

Each variant was trained for up to 40 epochs with the same optimizer, early-stopping rule, and $\mathrm{PE}_{\mathrm{clim}}$ evaluation as STORM-Bz. Table 3 summarizes mean PE over fifteen seeds. All three variants remain near $\mathrm{PE}_{1\mathrm{h}} = 0.987$ and $\mathrm{PE}_{6\mathrm{h}} \approx 0.91$. Pairwise differences among RDG, RDG-S, and SDG are within seed noise. None provides a consistent multi-horizon gain that would justify replacing the physics B_z gate as the default reported system. We therefore keep STORM-Bz as the primary model for interpretability (gate activation vs. B_z/Dst diagnostics) and treat RDG / RDG-S / SDG as evidence that short-horizon skill is largely a property of the shared training recipe rather than of one particular gate formula.

e: Repository name map.

For reproducibility, folder and `gate_type` strings in the code are unchanged: RDG $\leftrightarrow$ `cathode_anode` / `storm_cathode`; RDG-S $\leftrightarrow$ `storm_cathode_spec`; SDG $\leftrightarrow$ `radiotrophic` / `storm_radiotrophic`. Paper text uses only RDG, RDG-S, and SDG.

G. BASELINES AND TRAINING

We compare against Vanilla Transformer (same residual heads, no delay/gate), a Transformer baseline (default hyperparameters, not matched in width or depth to the STORM encoder), and LSTM. Training uses Adam [28] with initial learning rate 3×10^{-4}, cosine decay, batch size 64, and early stopping on 12 h validation MSE. Primary STORM and Transformer runs use seeds $\{42, \ldots, 56\}$. Transfer fine-tuning uses a lower learning rate for at most 50 epochs.

All models share the same preprocessor, the same chronological windows, and the same storm-biased sampler during training so that rare active intervals are not ignored by the optimizer. Early stopping monitors the 12 h validation MSE rather than training loss, which reduces the chance that a model is selected solely for fitting quiet-time persistence. Checkpoints used in the tables are the best validation PE weights for each seed; test PE is computed once after training and is not used for model selection.

We also train a Transformer with the same encoder hyperparameters as STORM ($d_{\mathrm{model}} = 128$, two layers, four heads; $\approx 1.19 \times 10^6$ parameters). This architecture-matched control is larger in parameter count than STORM and is not a parameter-matched baseline; the default Transformer uses $d_{\mathrm{model}} = 64$, three layers ($\approx 8.45 \times 10^5$ parameters). LSTM remains a useful sequential control because much of the older GEO flux literature is LSTM-based. We do not report a physics-augmented LSTM as a success in this manuscript: under the present trainer and loss schedule that variant failed to converge to competitive PE, and including it as a positive result would be misleading.

H. FEATURE ENGINEERING RATIONALE

Several of the sixteen solar-wind channels are derived rather than raw OMNI columns. Rolling means of V_{sw} and B_z give the encoder access to recent preconditioning without forcing the Transformer to learn long averaging filters from scratch. The product $B_z \times V_{sw}$ is a simple coupling proxy used in empirical radiation-belt work. Within-hour log-flux standard deviation preserves information that would otherwise be discarded when one-minute GOES samples are averaged to hourly means: a quiet hour and a turbulent hour can share a similar mean flux while carrying different short-timescale risk. Storm-onset hours and B_z-south duration summarize recent activity history in a form that is easy for a gate module to consume. None of these features is claimed to be novel in isolation; the point is that the released dataloader makes the inventory explicit so that ablations and transfer experiments remain reproducible.

V. RESULTS

A. GOES MULTI-SEED PERFORMANCE

On all-hour 6 h PE, the Transformer remains a strong baseline (0.904). A single STORM-Bz model is slightly lower (0.897; paired $p = 0.038$) but is significantly better at 1 h (0.986 versus 0.977, $p = 0.002$, Cohen's $d \approx 0.98$). The mixing weight α was selected on the validation split by sweeping $\alpha \in \{0.0, 0.3, 0.4, 0.5, 0.6, 0.7, 1.0\}$ and choosing the value that maximized validation $\mathrm{PE}_{6\mathrm{h}}$ ($\alpha^* = 0.3$). That fixed weight was then applied once to the held-out test set; the test-set curve in Fig. 3 is diagnostic only. At $\alpha^* = 0.3$, test PE reaches 0.911 at 6 h and 0.870 at 12 h (Table 4), above either single model. Bagging fifteen STORM seeds yields a similar multi-horizon profile without requiring the Transformer at inference; The short-STORM / long-TF hybrid recovers the Transformer's multi-horizon PE while retaining the STORM 1-hour score. The hybrid system routes the 1-hour head through a trained STORM-Bz model and the 6 h / 12 h heads

TABLE 4. GOES test PE against climatology (fifteen seeds unless noted). Brackets: bootstrap 95% CI where available. Bold: best primary system in column.

System	$\mathrm{PE}_{1\mathrm{h}}$	$\mathrm{PE}_{6\mathrm{h}}$	$\mathrm{PE}_{12\mathrm{h}}$	$\mathrm{PE}_{\mathrm{st},6\mathrm{h}}$
LSTM	0.961 $[0.957, 0.964]$	0.889 $[0.885, 0.892]$	0.853 $[0.851, 0.856]$	0.827 $[0.823, 0.831]$
Transformer	0.977 $[0.973, 0.981]$	0.904 $[0.901, 0.906]$	0.859 $[0.853, 0.863]$	0.821 $[0.814, 0.828]$
STORM-Bz	0.986 $[0.986, 0.987]$	0.897 $[0.894, 0.901]$	0.851 $[0.844, 0.853]$	0.827 $[0.821, 0.832]$
STORM (No-Delay)	0.987 $[0.987, 0.987]$	0.901 $[0.897, 0.903]$	0.853 $[0.848, 0.857]$	0.823 $[0.816, 0.830]$
STORM (No-Physics)	0.987 $[0.987, 0.987]$	0.901 $[0.898, 0.904]$	0.852 $[0.848, 0.855]$	0.824 $[0.816, 0.831]$
STORM (No-Gate)	0.987 $[0.986, 0.987]$	0.899 $[0.896, 0.902]$	0.853 $[0.848, 0.858]$	0.829 $[0.823, 0.834]$
Ensemble $\alpha^* = 0.3$[a]	0.983	0.911	0.870	0.839
Hybrid (short STORM / long TF)[a]	0.987	0.904	0.859	0.821
STORM bagged (15 seeds)	$\mathbf{0.988}$	0.911	0.870	$\mathbf{0.849}$
Transformer matched (mean)	0.981	0.907	0.861	—
Transformer matched (bagged)	0.986	$\mathbf{0.919}$	$\mathbf{0.877}$	—

Means over seeds 42–56. $\alpha^* = 0.3$ was selected on the validation split by maximizing $\mathrm{PE}_{6\mathrm{h}}$ and then applied once to the test set. Bagged rows average fifteen seed predictions before scoring (one aggregate evaluation). PE is $\mathrm{PE}_{\mathrm{clim}}$. Brackets are bootstrap 95% CIs where computed over seeds. [a]Ensemble and Hybrid rows are single aggregate evaluations (α^* applied once); bootstrap CIs are therefore not reported for these rows. Matched-Transformer rows are not bootstrapped because bagged evaluation uses a single aggregate prediction set.

through the corresponding Transformer model at inference time; no additional training is performed. Against persistence at 1 h, STORM remains near $+0.32$ while the Transformer is often negative (≈ -0.14).

Fig. 2 shows the same pattern seed-by-seed rather than as a single mean: the 1 h STORM advantage is stable across initializations.

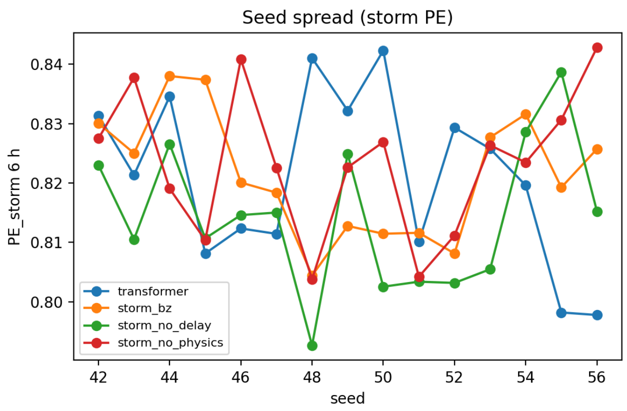

FIGURE 2. Seed-wise PE against climatology for Transformer and STORM-Bz (seeds 42–56). Short-horizon STORM scores cluster tightly above the Transformer; at 6 h the two families overlap, consistent with the paired tests in the text.

B. DST INTENSITY BINS

Table 5 stratifies the full GOES–OMNI working sample by physical Dst for STORM-Bz. Dst bins in Table 5 are a single-seed diagnostic and are not seed-averaged like Table 4. Skill is about 0.76 on quiet hours, rises to about 0.79–0.80 on minor and moderate storm bins, and remains about 0.71 on the sparse strong-storm bin ($\mathrm{Dst} \leq -100\,\mathrm{nT}$, $n = 206$). The model therefore does not rely only on quiet persistence-dominated intervals.

C. INPUT NOISE AND MISSING CHANNELS

Gaussian noise was added to standardized solar-wind inputs at test time ($\sigma \in \{0, 0.2, 0.5, 1.0, 1.5, 2.0\}$) for Transformer,

TABLE 5. STORM-Bz 6 h PE by Dst intensity[a]

Intensity bin	Count	$\mathrm{PE}_{6\mathrm{h}}$
Quiet ($\mathrm{Dst} > -30$)	77435	0.756
Minor ($-50 < \mathrm{Dst} \leq -30$)	7127	0.793
Moderate ($-100 < \mathrm{Dst} \leq -50$)	2316	0.800
Strong ($\mathrm{Dst} \leq -100$)	206	0.708

[a]Full chronological GOES–OMNI working sample (includes training data, so this is a descriptive diagnostic, not test-set PE); single-seed diagnostic, not averaged across seeds.

STORM-Bz, hybrid, and ensemble systems over fifteen seeds. At $\sigma = 2$, Transformer $\mathrm{PE}_{1\mathrm{h}}$ falls from about 0.96 to about 0.61, whereas STORM remains near 0.96. At 6 h the ensemble and STORM also lose less skill than the Transformer. Zeroing feature groups (no B_z; no velocity/density; no rolling averages) produces the same qualitative ordering (detailed tables provided in the accompanying code repository): STORM holds $\mathrm{PE}_{1\mathrm{h}}$ near 0.986–0.987 while the Transformer drops more when derived rolling channels are removed. These checks are controlled diagnostics, not a claim of immunity to arbitrary sensor failure.

D. COMPUTE COST

Table 6 reports parameter counts and batch-32 GPU latency in our timing setup. STORM-Bz is lighter than the default Transformer baseline while remaining competitive on all-hour 6 h PE and improving short-horizon reliability. An architecture-matched Transformer control ($d_{\mathrm{model}} = 128$, 2 layers, 4 heads; $\approx 1.19 \times 10^6$ parameters) is reported separately in Table 4.

TABLE 6. Model size and inference cost (batch 32, GPU, single seed)

Model	Parameters	ms / batch
Transformer (baseline)	8.45×10^5	2.95
STORM-Bz	3.43×10^5	3.73

The delay-module interpolation introduces a sequential

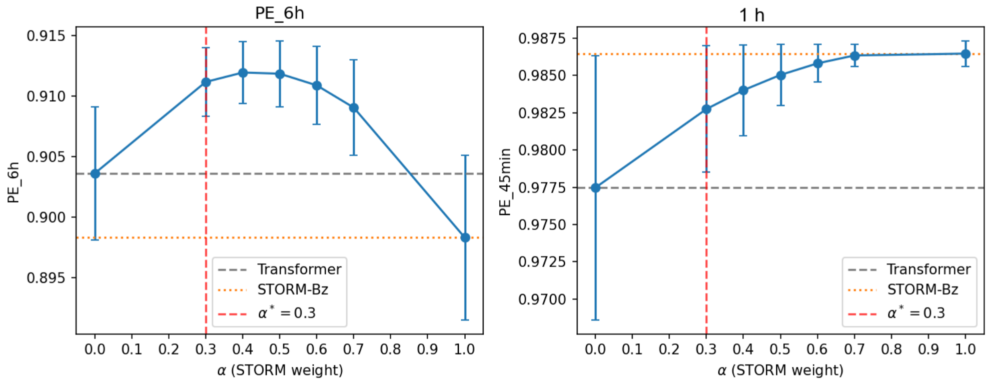

FIGURE 3. Validation PE_{6h} versus mixing weight α. Vertical line: $\alpha^* = 0.3$ selected on validation; corresponding test PE is reported in Table 4.

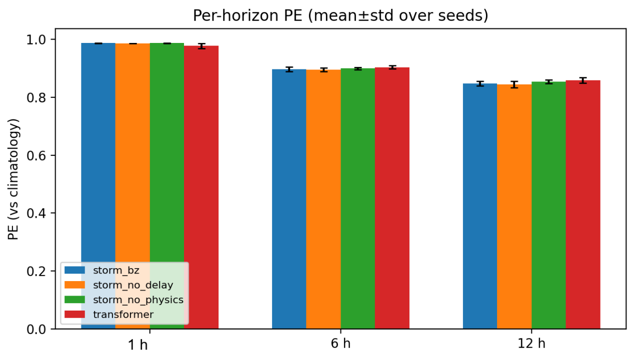

FIGURE 4. Per-horizon PE against climatology (mean±std, fifteen seeds). STORM-Bz leads at 1 h; at 6 h and 12 h the single models are close, while the ensemble improves multi-horizon PE (see Table 4).

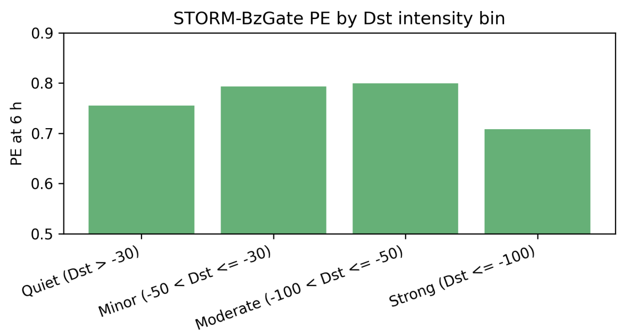

FIGURE 5. STORM-Bz 6 h PE stratified by Dst intensity bin on the full GOES–OMNI working sample. **Note: this includes training data and is a descriptive diagnostic, not test-set PE.**

operation not present in the pure Transformer, which accounts for the modest per-batch latency increase despite the smaller parameter count.

E. EVENT TIME SERIES

Fig. 9 shows STORM-Bz predictions against observed log-flux during a major flux enhancement event in the GOES test period. The model captures the broad shape of the multi-day enhancement and recovery more closely than persistence. We do not claim the gate panel tracks reconnection from these plots alone. The model struggles with sharp dropout and quiet conditions (Fig. 11 and Fig. 10), which are included as failure modes rather than success cases.

F. RESIDUAL STRUCTURE

Residual histograms (Figs. 7–8) show that errors concentrate near zero on quiet hours and widen in storm windows for both models; STORM does not eliminate storm-time residual variance, but it reduces the short-horizon bias relative to persistence discussed above.

G. INTERPRETABILITY

Fig. 14 relates the two trainable physics modules to familiar solar-wind drivers; these panels are diagnostic, not a claim that τ recovers a unique physical constant on every window.

H. GRASP TRANSFER

On GSAT-19 GRASP (fifteen seeds), zero-shot GOES models already achieve short-horizon PE near 0.82, but 6 h and 12 h PE fall. Fine-tuning on GRASP (freezing the Transformer encoder and updating only the forecast heads) raises 6 h PE from 0.449 to 0.599 (paired t-test across fifteen seeds, $p = 7.6 \times 10^{-7}$, Cohen's $d = 2.17$) and 12 h PE from 0.182 to 0.517 ($p = 3.6 \times 10^{-10}$, $d = 3.98$). The inter-seed standard deviation after fine-tuning remains 0.073 at both 6 h and 12 h, indicating that transfer convergence is still seed-sensitive; future work should examine learning-rate schedules specific to the fine-tuning stage. Absolute GRASP PE remains below the GOES test scores, as expected under sensor and longitude shift; the practical point is the large mid- and long-horizon recovery after adaptation rather than a claim of parity with GOES.

I. HORIZON-RESTRICTED PHYSICS LOSS

We also trained a variant that applies physics-loss terms only on the short head (ignoring the 6 h and 12 h heads). Across fifteen seeds, this variant did not differ significantly from the baseline STORM model on PE_{6h} (paired $p \approx 0.64$). We

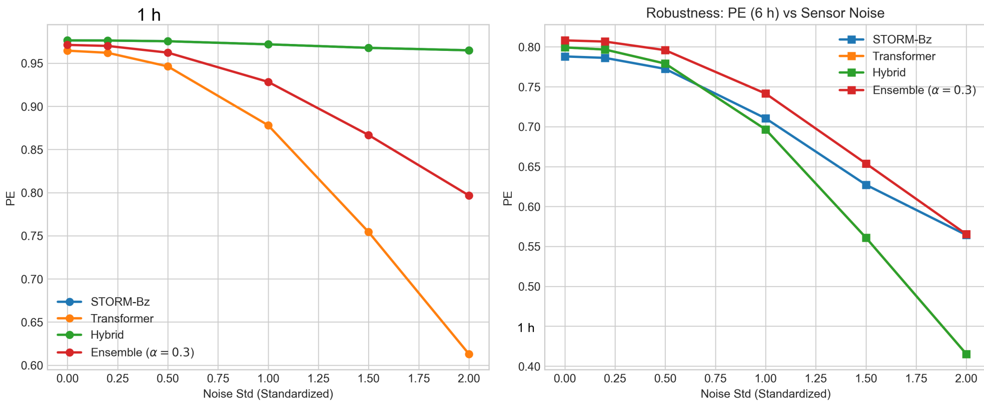

FIGURE 6. Prediction efficiency under Gaussian noise on standardized solar-wind inputs (fifteen seeds). STORM and hybrid retain short-horizon PE under strong noise; the Transformer degrades sharply at 1 h. Ensemble remains competitive at moderate noise levels.

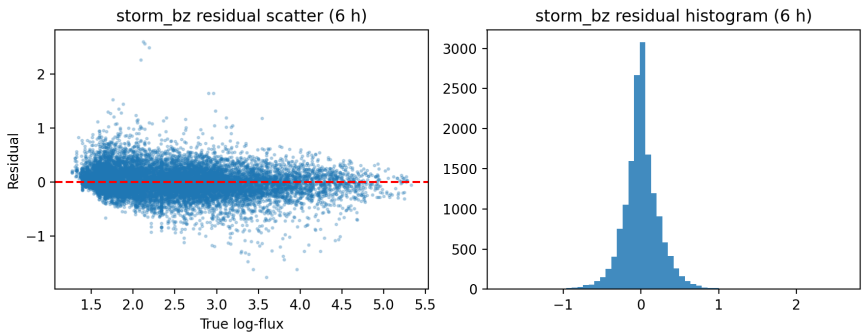

FIGURE 7. Residual distribution (predicted minus true log-flux) for STORM-Bz on the GOES test set, stratified by storm and quiet windows.

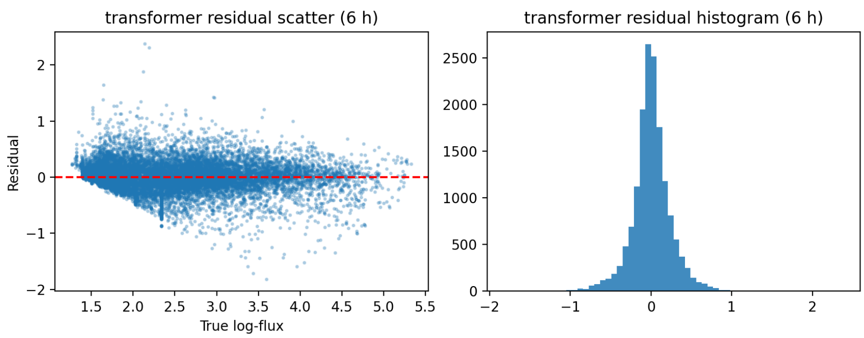

FIGURE 8. Same residual view for the Transformer baseline.

TABLE 7. GRASP (GSAT-19) transfer results (fifteen seeds). Paired tests across fifteen seeds are reported in the text. Bold: best per horizon.

Horizon	Zero-shot	Fine-tuned	Gain
1 h	0.820 ± 0.001	0.827 ± 0.009	+0.007
6 h	0.449 ± 0.011	$\mathbf{0.599 \pm 0.073}$	**+0.150**
12 h	0.182 ± 0.045	$\mathbf{0.517 \pm 0.073}$	**+0.335**

therefore treat horizon-restricted physics weighting as a null ablation rather than a performance lever.

We further trained a No-Gate variant that sets the modulation to identity ($g = 1$). On $\mathrm{PE}_{\mathrm{clim}}$, removing the gate does not reduce 1 h skill relative to full STORM-Bz (0.987 versus 0.986 in Table 4); the same null result holds for the No-Delay and No-Physics ablations. The short-horizon gain versus the Transformer is therefore a property of the trained STORM package under our protocol, not a gain that disappears when any one physics module is removed. We do not attribute that

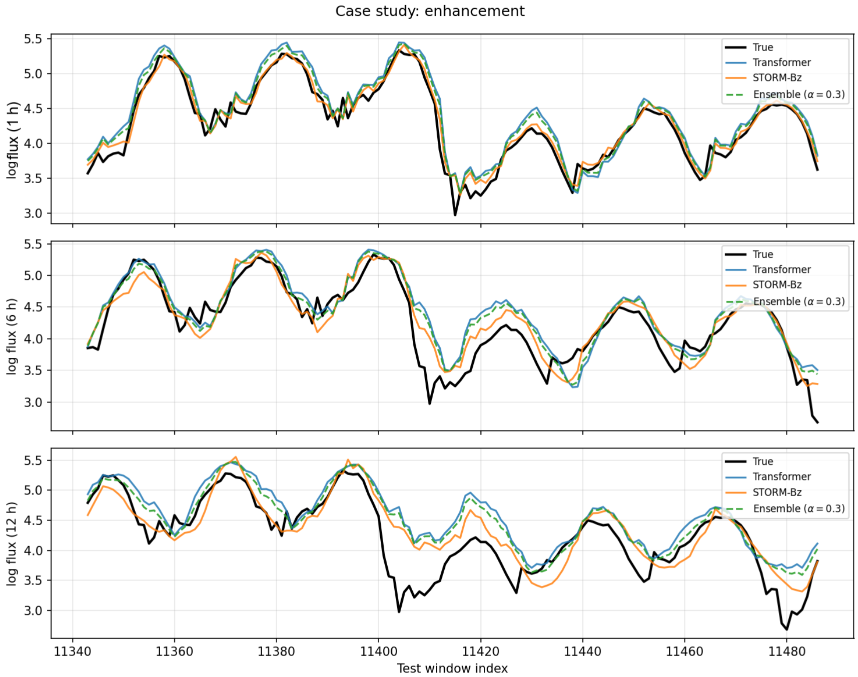

FIGURE 9. Illustrative enhancement window. Top: true log-flux, STORM-Bz, and persistence. Lower panels: Bz, Vsw, predicted τ, and gate. τ is constrained to $[0.5, 1.5]$ h.

FIGURE 10. Quiet drift failure mode. The model overpredicts gradual flux recovery during low geomagnetic activity.

1 h $\mathrm{PE}_{\mathrm{clim}}$ edge to the gate alone.

On storm windows, single-model $\mathrm{PE}_{\mathrm{st,6h}}$ values are close: STORM-Bz 0.827, LSTM 0.827, Transformer 0.821, with No-Delay and No-Physics slightly lower (0.823 and 0.824) and No-Gate comparable or slightly higher (0.829). Given overlapping intervals, we do not claim that any one physics module is required for storm-time skill. The clearest gains on storm subsets remain post-hoc aggregation: the $\alpha^* = 0.3$ ensemble reaches 0.839 and seed-bagged STORM reaches 0.849 (Table 4).

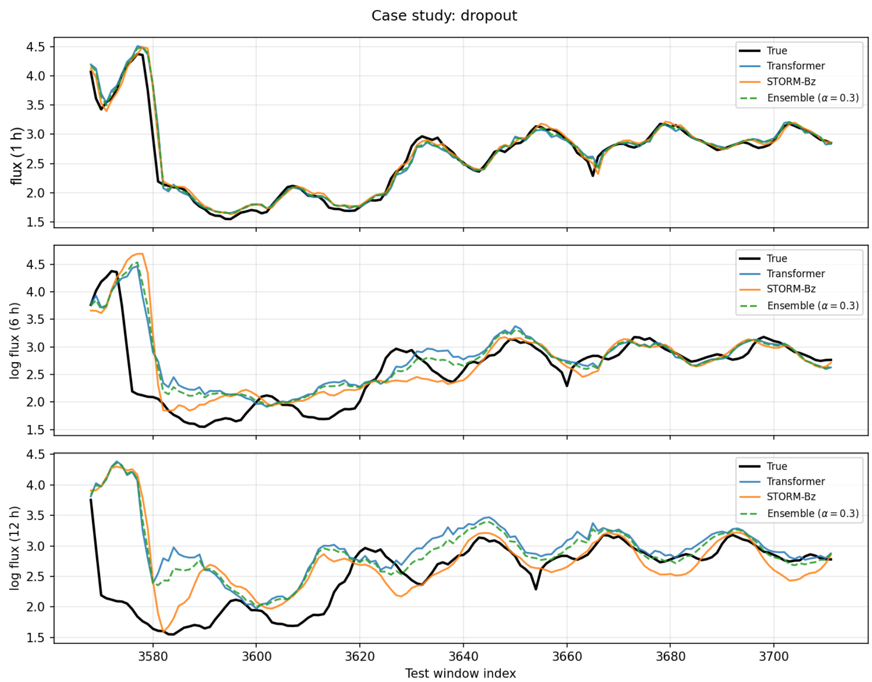

FIGURE 11. Sharp dropout failure mode. The model does not fully capture the rapid magnetopause shadowing flux drop.

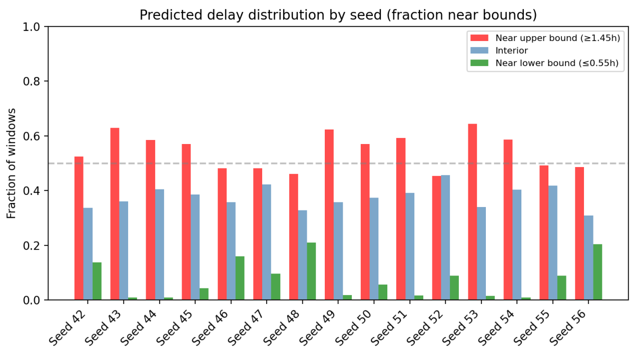

FIGURE 12. Fraction of predicted propagation delays τ near the constraint bounds across fifteen seeds. The blue bars show the interior fraction, red bars show the fraction near the upper bound ($1.5\,\mathrm{h}$), and green bars show the fraction near the lower bound ($0.5\,\mathrm{h}$). The saturation at the upper bound indicates the box constraint is active.

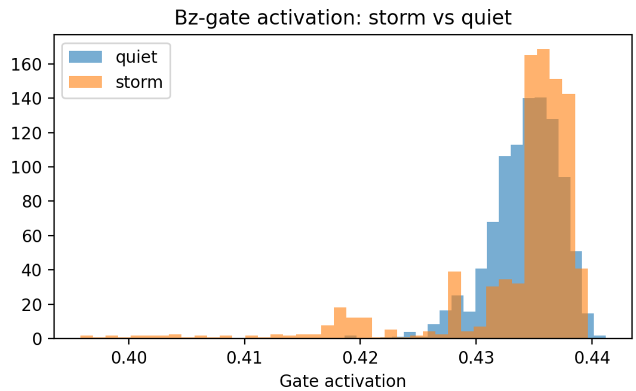

FIGURE 13. B_z gate activation: storm windows (orange) versus quiet intervals (blue) (Mann–Whitney $p = 1.2 \times 10^{-14}$).

Fig. 16 summarizes the same controlled comparison at 6 h: ablating delay or physics loss does not overturn the ranking already reported in Table 4.

VI. DISCUSSION

The controlled comparison supports a layered reading rather than a single "winner" architecture. At 1 h, STORM improves reliability relative to a strong Transformer, especially against persistence. Controlled ablations that remove the delay, the physics loss, or the gate do not by themselves erase that short-horizon $\mathrm{PE}_{\mathrm{clim}}$ edge. We therefore treat the gain as a property of the full trained system rather than as proof that any single module is necessary.

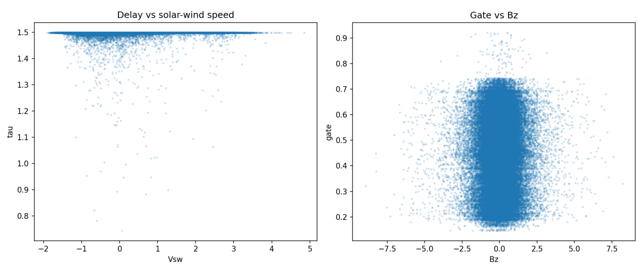

FIGURE 14. Physics-module diagnostics on the GOES test set: predicted delay τ against solar-wind speed, and gate activation against southward B_z statistics. Delays remain inside the imposed $[0.5, 1.5]\,\mathrm{h}$ band; gate values rise when southward IMF is sustained.

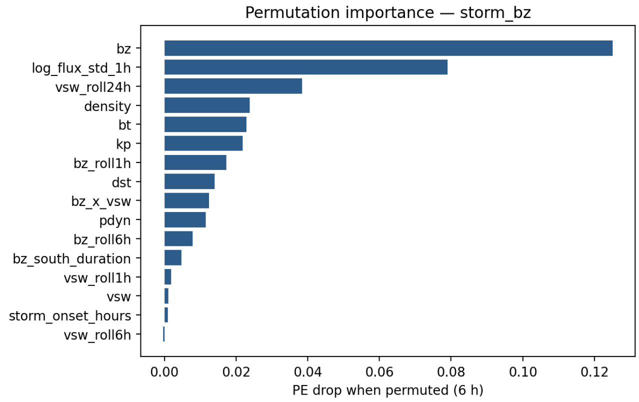

FIGURE 15. Permutation importance on the $6\,\mathrm{h}$ head for STORM-Bz (PE drop when each input channel is shuffled). B_z and related solar-wind drivers rank highest.

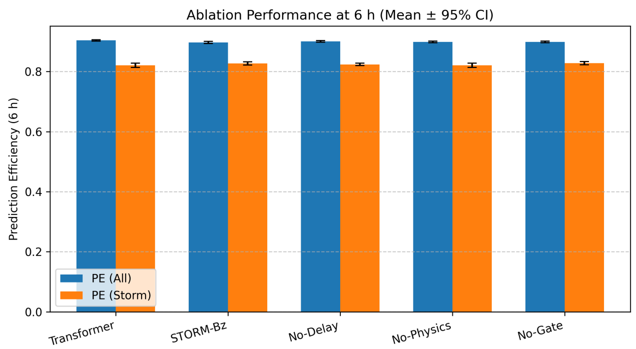

FIGURE 16. Six-hour PE against climatology for the Transformer baseline, full STORM-Bz, and ablations that remove the delay module or the physics-loss term (fifteen seeds; mean$\pm$std). Removing the delay module, the physics-loss term, or the B_z gate does not collapse $\mathrm{PE_{6h}}$ or remove the 1 h advantage over the Transformer; the main single-model STORM advantage remains the short-horizon comparison in Table 4.

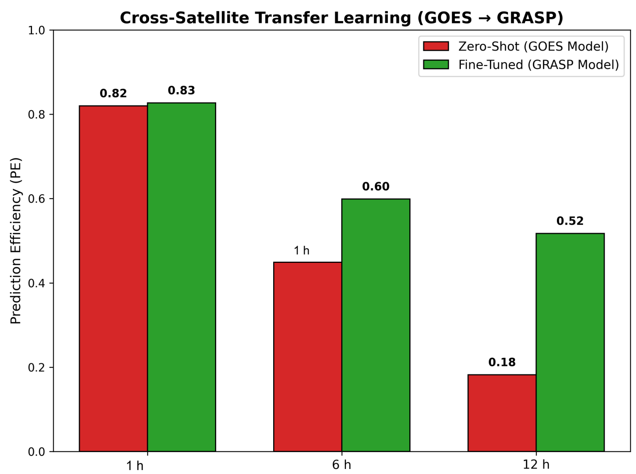

FIGURE 17. GRASP domain-gap summary (fifteen seeds): zero-shot GOES models versus fine-tuning on GSAT-19. Short-horizon PE transfers more readily; 6 h and 12 h PE improve substantially after adaptation.

As detailed in Section V, the ablations together indicate that the short-horizon gain is produced by the overall training protocol rather than by any single runtime module. The delay and gate act as structured inductive biases during training; once the model is trained, removing either module at inference does not erase the 1 h $\mathrm{PE_{clim}}$ edge over the Transformer. STORM-PhysNet should therefore be understood primarily as a carefully engineered training and multi-horizon recipe that includes interpretable physics-inspired components. To test whether the original upper bound of $1.5\,\mathrm{h}$ was limiting performance, we retrained STORM with wider delay constraints (2.0, 2.5, 3.0, 3.5, and $4.0\,\mathrm{h}$) across fifteen seeds (sixteen seeds for the $2.0\,\mathrm{h}$ bound). Mean $\mathrm{PE_{1h}}$ remained in the narrow range 0.9859–0.9862 and $\mathrm{PE_{6h}}$ remained in the range 0.900–0.902 for all bounds (see Fig. 18). The original $[0.5, 1.5]\,\mathrm{h}$ constraint is therefore not a performance bottleneck.

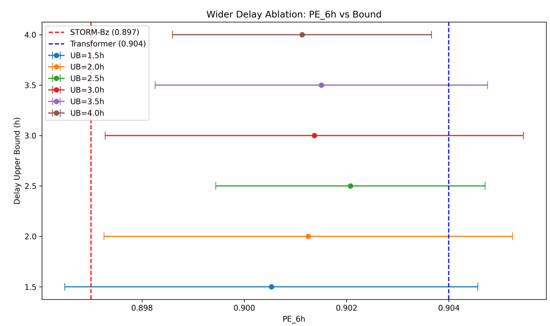

FIGURE 18. Wider delay ablation: mean $\mathrm{PE_{6h}}$ versus delay upper bound (error bars show seed standard deviation). Vertical dashed lines mark the paper baselines STORM-Bz (0.897) and Transformer (0.904). $\mathrm{PE_{6h}}$ remains stable across upper bounds from $1.5\,\mathrm{h}$ to $4.0\,\mathrm{h}$, confirming that the original $[0.5, 1.5]\,\mathrm{h}$ constraint is not a performance bottleneck.

An architecture-matched Transformer control ($d_{\mathrm{model}} = 128$, two layers) reached mean $\mathrm{PE_{1h}} = 0.981$ and $\mathrm{PE_{6h}} = 0.907$ across fifteen seeds; bagging the fifteen matched seeds yields $\mathrm{PE_{1h}} = 0.986$, $\mathrm{PE_{6h}} = 0.919$, and $\mathrm{PE_{12h}} = 0.877$. Relative to STORM-Bz ($0.986$ / 0.897 / 0.851), matching encoder width/depth shrinks the short-horizon gap and shifts the multi-horizon advantage toward the larger Transformer under bagging. This reinforces that single-model PE gaps are protocol-plus-width/depth effects rather than proof that the delay or gate modules are individually necessary.

At 6–12 h, a plain Transformer remains competitive as a single model; combining the two predictors (or bagging STORM seeds) provides the clearest multi-horizon gain among the systems evaluated here. Seed-bagged STORM reaches a multi-horizon PE level comparable to the linear ensemble (0.911 at 6 h). Noise and missing-channel probes indicate that the physics-inspired model fails more gracefully when upstream solar-wind features are corrupted. GRASP results show that Indian-longitude deployment still needs local adaptation for longer leads, but short-horizon skill transfers more readily than long-horizon skill.

Dst intensity bins reinforce that reading. Skill does not vanish in moderate or strong storm bins; the strong-storm bin is small ($n = 206$), so uncertainty is higher, but PE remains clearly above zero. Robustness grids show a smooth degradation under noise rather than a brittle cliff at the first perturbation, which is desirable for a research prototype even though it is not a substitute for hardware-in-the-loop testing.

A. TRANSFER LEARNING FOR INDIAN LONGITUDE

The GRASP experiments illustrate a practical constraint for new GEO instruments: a GOES-pretrained model can retain useful short-horizon skill zero-shot, but mid- and long-horizon PE drop until local fine-tuning is applied. In our fifteen-seed runs, fine-tuning raises 6 h PE from about 0.45 to about 0.60 and 12 h PE from about 0.18 to about 0.52. Absolute GRASP PE remains below the GOES test scores, as expected under sensor and longitude shift. A similar adaptation profile is observed for the architecture-matched Transformer baseline, which improves its 6 h PE from 0.549 ± 0.015 (zero-shot) to 0.625 ± 0.021 after fine-tuning.

B. COMPARABILITY WITH PUBLISHED PE NUMBERS

PE values reported across studies should not be compared without normalization for data period, solar-cycle phase, energy channel, cadence, and persistence baseline [21], [22]. Daily fluence scores near 0.9 on one archive are not evidence that an hourly log-flux model on 2012–2016 is "behind." Our training period spans a moderate activity phase of Solar Cycle 24; results during a more active phase may differ. The contribution of this paper is a controlled multi-seed comparison and a documented transfer recipe.

VII. LIMITATIONS

All reported leads are integer hours on hourly data; sub-hour (e.g. 45 min) supervision would require higher-cadence flux labels. Main GOES tables use fifteen seeds with bootstrap intervals; some secondary diagnostics (individual case windows, older single-seed robustness grids) remain thinner. PE values are not interchangeable with daily-fluence scores reported elsewhere without matching cadence, energy channel, and baseline [21], [22]. GRASP covers a limited declining-phase window. Performance during solar maximum (Cycle 25, 2023–2025) is unknown and likely to differ, particularly in the storm-time bins. Ensemble skill uses a fixed $\alpha^* = 0.3$ selected on the validation split; the test-set α curve is diagnostic only. We do not claim online α adaptation. Horizon-restricted physics losses did not improve PE in our tests. The reported p-values are uncorrected for multiple comparisons across horizons and ablations; treat individual significance claims as descriptive rather than confirmatory. The default Transformer baseline is not architecture-matched to STORM; the matched control ($d_{\mathrm{model}} = 128$, two layers) shows that encoder width/depth and bagging account for much of the multi-horizon PE movement, so we still do not attribute single-model gaps solely to the delay or gate modules.

Alternative gates, a hybrid SSM backbone, magnetopause features, and a spectral head were implemented and briefly evaluated; they are not used in the reported tables (`backbone=transformer`, spectral head off).

VIII. CONCLUSION

STORM-PhysNet is a reproducible multi-horizon baseline on GOES with a documented transfer path toward GSAT-19 GRASP. Relative to a strong Transformer, the clearest single-model gain is short-horizon reliability (including a large gap against persistence at 1 h). The clearest multi-horizon gain among the systems evaluated here comes from an architecture-matched bagged Transformer, which reaches test $\mathrm{PE}_{6\mathrm{h}} = 0.919$ and $\mathrm{PE}_{12\mathrm{h}} = 0.877$. A linear ensemble with validation-selected $\alpha^* = 0.3$ and seed-bagged STORM also achieve strong multi-horizon results (both reaching $\mathrm{PE}_{6\mathrm{h}} = 0.911$). Input-noise and missing-channel checks, together with GRASP fine-tuning gains at 6–12 h, support operational motivation beyond a pure leaderboard PE number; for instance, the architecture-matched baseline achieves a marginal persistence prediction efficiency ($\mathrm{PE}_{\mathrm{pers}}$) of 0.053 ± 0.148 at 1 h, illustrating the difficulty of solidly outperforming operational persistence models. Furthermore, under synthetic Gaussian noise injection ($\sigma = 1.0$), the 6 h PE of the matched baseline drops to 0.825 ± 0.011, highlighting the need for structurally robust architectures during degraded sensor conditions. Horizon-restricted physics losses did not improve PE in a fifteen-seed ablation.

DATA AVAILABILITY

The code, configuration files, trained checkpoints, evaluation scripts, result CSVs, paper figures, and the chronological split protocol are available at https://github.com/bnsama29-cloud/STORM-PhysNet. The GOES-15 EPEAD dataset is publicly available from NOAA NCEI (https://www.ngdc.noaa.gov/stp/satellite/goes/). OMNI is available via NASA OMNIWeb (https://omniweb.gsfc.nasa.gov/). GSAT-19 GRASP is accessible via ISSDC (https://www.issdc.gov.in/).

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REFERENCES

[1] D. N. Baker et al., "Space weather effects in the Earth's radiation belts," Space Sci. Rev., vol. 214, no. 1, p. 17, 2018.
[2] R. B. Horne et al., "Wave acceleration of electrons in the Van Allen radiation belts," Nature, vol. 437, pp. 227–230, 2005.
[3] D. L. Turner et al., "Explaining sudden losses of outer radiation belt electrons during geomagnetic storms," Nature Phys., vol. 8, pp. 208–212, 2012.
[4] D. N. Baker et al., "Linear prediction filter analysis of relativistic electron properties at 6.6 R_E," J. Geophys. Res., vol. 95, pp. 15133–15140, 1990.
[5] K. A. Perry et al., "Comparing geosynchronous relativistic electron prediction models," Space Weather, vol. 8, S12003, 2010.
[6] J. Bortnik and R. M. Thorne, "The dual role of ELF/VLF chorus waves in the acceleration and precipitation of radiation belt electrons," J. Atmos. Sol.-Terr. Phys., vol. 69, pp. 378–386, 2007.
[7] A. G. Ling et al., "Identifying solar wind drivers of radiation belt responses," Space Weather, vol. 8, no. 10, 2010.

[8] L. Wei et al., “Quantitative prediction of high-energy electron integral flux at geostationary orbit based on deep learning,” Space Weather, vol. 16, pp. 903–916, 2018.
[9] J. Son et al., “72-hour time series forecasting of hourly relativistic electron fluxes at geostationary orbit by deep learning,” Space Weather, vol. 20, e2022SW003153, 2022.
[10] H. Zhang et al., “A prediction model of relativistic electrons at geostationary orbit using the EMD-LSTM network,” Space Weather, vol. 20, e2022SW003126, 2022.
[11] L. E. Simms et al., “Classifier neural network models predict relativistic electron events at geosynchronous orbit,” J. Geophys. Res. Space Physics, vol. 125, e2019JA027323, 2020.
[12] Y. Wang et al., “LSTM-based modeling of radiation belt electron dynamics using MagEIS Van Allen Probes observations,” Space Weather, vol. 23, e2024SW004001, 2025.
[13] A. Vaswani et al., “Attention is all you need,” in Proc. NeurIPS, 2017, pp. 5998–6008.
[14] W. Tan et al., “Comparative analysis of TPA-LSTM and Transformer models for forecasting GEO radiation belt electron fluxes,” Space Weather, vol. 22, e2023SW003714, 2024.
[15] X. Sun et al., “A modeling study of $\geq$ 2 MeV electron fluxes in GEO at different prediction time scales based on LSTM and transformer networks,” J. Space Weather Space Clim., vol. 14, p. 20, 2024.
[16] X. Sun et al., “Enhancing $\geq$ 2 MeV electron fluence predictions in GEO through three deep learning strategies,” Space Weather, vol. 23, e2024SW003975, 2025.
[17] G. D. Reeves et al., “Acceleration and loss of relativistic electrons during geomagnetic storms,” Geophys. Res. Lett., vol. 30, no. 10, 2003.
[18] R. M. Thorne, “Radiation belt dynamics: The importance of wave-particle interactions,” Geophys. Res. Lett., vol. 37, L22107, 2010.
[19] T. P. O’Brien et al., “Which magnetic storms produce relativistic electrons at geosynchronous orbit?” J. Geophys. Res., vol. 106, pp. 15533–15544, 2001.
[20] Y. Y. Shprits et al., “Review of modeling of losses and sources of relativistic electrons in the outer radiation belt,” J. Atmos. Sol.-Terr. Phys., vol. 70, pp. 1694–1713, 2008.
[21] E. Camporeale, “The challenge of machine learning in space weather: Nowcasting and forecasting,” Space Weather, vol. 17, pp. 1166–1207, 2019.
[22] S. K. Morley et al., “Evaluating the performance of radiation belt models: Metrics, baselines, and statistics,” Space Weather, vol. 16, pp. 1521–1548, 2018.
[23] R. Tang et al., “Short-time prediction of energetic electron flux based on stacking ensemble learning,” Space Weather, vol. 20, e2021SW002928, 2022.
[24] Y. Liu et al., “iTransformer: Inverted Transformers are effective for time series forecasting,” in Proc. ICLR 2024 (Spotlight), 2024. arXiv:2310.06625.
[25] S. Sinha et al., “PreMevE update: Forecasting ultra-relativistic electrons inside Earth’s outer radiation belt,” Space Weather, vol. 19, e2021SW002773, 2021.
[26] E. Botek et al., “Prediction of radiation belt electron fluxes at LEO using neural networks with PROBA-V/EPT data,” Space Weather, vol. 21, e2023SW003466, 2023.
[27] N. Y. Ganushkina et al., “Forecasting GOES 15 $>$ 2 MeV electron fluxes from solar wind data and geomagnetic indices,” Space Weather, vol. 18, e2019SW002416, 2020.
[28] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in Proc. Int. Conf. Learn. Represent. (ICLR), 2015.

Samarth BN is pursuing the B.E. degree in Electronics and Communication Engineering at RV College of Engineering, Bengaluru, India. His interests include space-weather forecasting, time-series machine learning, and radiation-belt physics.